

Sculpt3D — Requirements Analysis

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Contents

1	Introduction	2
1.1	Evidence of the accessibility gap	2
2	Stakeholders and target users	3
3	Scope	4
4	Functional requirements	4
4.1	Scene and general	5
4.2	Sculpting	5
4.3	Polygon drawing and extrusion	5
4.4	Physics	6
4.5	Stretch modules	6
5	Educational requirements	6
6	Non-functional requirements	7
7	Key use cases	7
8	Constraints and assumptions	8
9	Prioritization and release plan	8

1 Introduction

Sculpt3D (name is not final) is a desktop 3D modelling tool that trades raw power and a little “freedom” for a much shorter learning curve and an intuitive UI: a moderately capable tool anyone can pick up in minutes, plus built-in teaching of the geometry and physics behind it.

Problem. Existing 3D tools sit at two extremes. Professional suites (Blender, Maya, ZBrush, Rhino) are powerful but take weeks to become productive in. Toy-level apps are easy but too limited to produce anything usable. Between them there is a gap: users who need to get a real job done quickly, without becoming 3D specialists.

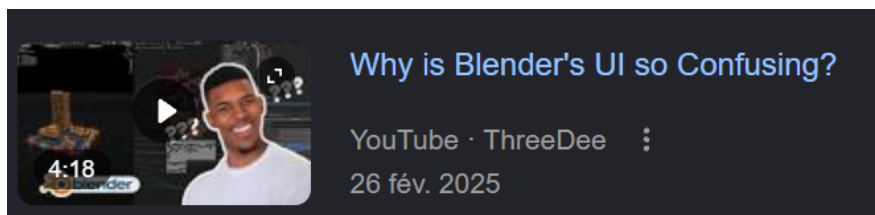
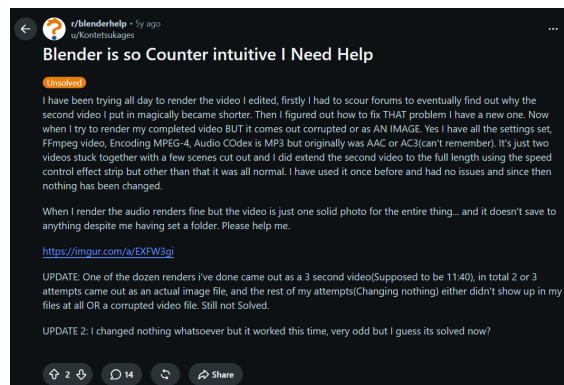
Second gap: education. There are few accessible tools that let a student *see* the concepts behind 3D work — vertices and normals, extrusion, transformations, forces and motion. Teaching material and modelling software are usually separate things.

Vision. One tool covering four connected activities: sculpt shapes, draw and extrude polygons, run basic mechanics on what you built, and (later) animate it. Every feature should be usable without a manual, and every feature should be able to explain itself.

Guiding principle. When power and accessibility conflict, accessibility wins. A feature that needs a tutorial to discover is redesigned, not documented.

1.1 Evidence of the accessibility gap

The learning-curve problem is widely reported by users themselves. Even favourable reviews of Blender, the leading free tool, repeatedly mention a steep learning curve, a complex interface, and too many tools and shortcuts to remember [1]. Industry training providers make the same point, contrasting Blender’s confusing interface with more beginner-friendly packages [2]. Exhibits ??-?? show typical beginner experiences.





2 Stakeholders and target users

Four user groups drive the requirements; the first two are primary.

Persona	Who	Goal	Pain today	Needs from Sculpt3D
Architecture student (primary)	1st–3rd year, some CAD exposure	Quick massing models and form studies	Rhino/Revit too heavy for early sketches	Draw a footprint, extrude it, sculpt, export
Learner / teacher (primary)	High school or early university, physics or geometry classes	Understand 3D geometry and basic mechanics visually	Concepts taught on paper; tools don't explain themselves	Guided lessons, visible vectors and forces, step-by-step mode
3D enthusiast / hobbyist	No formal training	Make shapes for fun, games, printing	Blender's learning curve	Instant sculpting, undo, simple export
Architect (secondary)	Practising professional	Fast concept sketches before real CAD	Concept-stage tools are slow to start	Clean polygon drawing, dimensions, export to their main tool

Table 1: Target user personas.

Other stakeholders: course supervisors (evaluate the project), the development team (maintain it), and potentially schools or teachers who adopt it as a classroom tool.

3 Scope

The project ships three core modules, with animation and fluid simulation as stretch goals only if the core is solid.

Module	Status	Summary
Sculpting	Core (prototype exists)	Push, pull, smooth and shape meshes with brushes
Polygon drawing + extrusion	Core	Draw a flat polygon, then extrude it into a solid with a separate tool
Physics (basic mechanics)	Core	Gravity, mass, collisions, forces and simple rigid-body motion on created objects
Education layer	Core	Lessons, explanations and visual overlays tied to the modules above
Animation	Stretch	Keyframe transforms over a timeline, playback
Fluid simulation	Stretch	Simple particle-based liquid for demonstration, not engineering accuracy

Table 2: Project modules.

Explicitly out of scope: photorealistic rendering, texturing and UV mapping, CAD-grade precision (tolerances, BIM data), multi-user collaboration, mobile and web versions, scripting/plugin API, and engineering-accurate simulation.

4 Functional requirements

Each requirement has an ID for traceability and a MoSCoW priority (M = Must, S = Should, C = Could).

4.1 Scene and general

ID	Requirement	Priority
FR-G1	Create, select, rename, duplicate and delete objects from an object list	M
FR-G2	Orbit, pan and zoom the camera with the mouse; reset view with one key	M
FR-G3	Undo/redo for every editing action	M
FR-G4	Insert primitives (cube, sphere, cylinder, plane)	M
FR-G5	Move, rotate and scale objects with on-screen handles	M
FR-G6	Save and load projects in a native format	M
FR-G7	Export to OBJ and STL (for 3D printing and other tools)	M
FR-G8	Import OBJ	S
FR-G9	Grid, snapping and an optional dimension readout in metres	S

Table 3: Scene and general requirements.

4.2 Sculpting

ID	Requirement	Priority
FR-S1	Brushes: pull/push, smooth, flatten, pinch	M
FR-S2	Adjustable brush radius and strength via sliders and shortcuts	M
FR-S3	Brushes preserve the overall shape (no mesh tearing or spikes)	M
FR-S4	Symmetry mode across X, Y or Z	S
FR-S5	Mesh subdivision on demand for finer detail	S

Table 4: Sculpting requirements.

4.3 Polygon drawing and extrusion

ID	Requirement	Priority
FR-D1	Draw tool: trace a flat polygon by clicking points on a plane; close it by clicking the first point	M
FR-D2	Freehand mode: drag to draw, path simplified into a clean polygon	S
FR-D3	Extrude tool, separate from Draw: select a polygon and push it out perpendicular to its plane	M
FR-D4	Extrusion height set by dragging or typed value	M
FR-D5	Edit polygon vertices after drawing	S
FR-D6	Draw on any face of an existing object, not just the ground plane	C

Table 5: Polygon drawing and extrusion requirements.

4.4 Physics

ID	Requirement	Priority
FR-P1	Play/pause/step/reset simulation; reset restores pre-simulation state	M
FR-P2	Per-object properties: mass, static/dynamic, bounciness, friction	M
FR-P3	Gravity with adjustable strength and direction	M
FR-P4	Collisions between objects and with the ground	M
FR-P5	Apply a force or impulse to an object by click-and-drag	S
FR-P6	Live readout of velocity, acceleration and energy for a selected object	S
FR-P7	Slow-motion and frame-by-frame stepping	S
FR-P8	Springs and simple joints between objects	C

Table 6: Physics requirements.

4.5 Stretch modules

ID	Requirement	Priority
FR-A1	Keyframe position, rotation and scale on a timeline; play back	C
FR-A2	Record a physics run as an animation	C
FR-F1	Particle-based fluid emitter interacting with scene objects	C

Table 7: Stretch module requirements.

5 Educational requirements

The education layer is what separates Sculpt3D from other simple modellers: the tool teaches the concepts it is built on, inside the same workspace.

ID	Requirement	Priority
ED-1	“Explain” toggle: hovering a tool shows what it does and the concept behind it (e.g. extrusion = sweeping a 2D shape along its normal)	M
ED-2	Visual overlays on demand: vertices, edges, face normals, local axes	M
ED-3	Physics overlays: force arrows, velocity vectors, trajectory trail, centre of mass	M
ED-4	Guided lessons: short interactive tutorials (5–10 min) that check progress in the scene, e.g. “extrude this footprint to 3m”	M
ED-5	Starter lesson set: navigation, sculpting basics, polygons and extrusion, gravity and free fall, collisions and energy, projectile motion	S
ED-6	Experiment mode: change one variable (mass, gravity, angle) and compare runs side by side or as overlaid trails	S
ED-7	Formula panel: show the equation being simulated with live values plugged in (e.g. $v = v_0 + g t$)	S
ED-8	Teacher can save a scene as a ready-made exercise and share the file with students	C
ED-9	Interface available in English and French; Arabic as a later addition	C

Table 8: Educational requirements.

6 Non-functional requirements

Accessibility is the headline feature, so it is stated as measurable targets rather than a slogan.

ID	Category	Requirement
NFR-1	Learnability	A first-time user creates and exports a simple shape within 10 minutes, with no external documentation
NFR-2	Learnability	Every tool is reachable in at most 2 clicks from the main screen; no nested menus deeper than 2 levels
NFR-3	Usability	At most ~ 12 tools visible by default; advanced options hidden behind an “Advanced” expander
NFR-4	Usability	Every tool shows a one-line hint in a status bar when selected
NFR-5	Accessibility	Full keyboard shortcuts for common actions; adjustable UI scale (100–200%); colour-blind-safe palette for overlays and axes
NFR-6	Forgiveness	Every action undoable; autosave every 2 minutes; no destructive action without confirmation
NFR-7	Performance	Viewport stays at 30+ FPS on a mid-range laptop with integrated graphics for meshes up to $\sim 100k$ triangles
NFR-8	Performance	Physics runs in real time for 50 rigid bodies; brush strokes respond in under 50 ms
NFR-9	Portability	Runs on Windows, Linux and macOS (Java/JavaFX)
NFR-10	Installability	Single installer or self-contained bundle; no separate Java install required
NFR-11	Cost	Free to use for students and schools
NFR-12	Maintainability	Modules (sculpt, draw, physics, education) are separate packages with clear interfaces so stretch modules plug in without rewrites
NFR-13	Reliability	No data loss on crash: recovery from latest autosave on next launch

Table 9: Non-functional requirements.

7 Key use cases

Five use cases cover the main flows; each maps to the requirements above.

UC-1 — Massing model (architecture student)

Draws a building footprint on the ground plane (FR-D1), extrudes it to 12 m (FR-D3, FR-D4), adds a second extruded volume, sculpts a curved roof (FR-S1), exports to OBJ for their main CAD tool (FR-G7).

UC-2 — Free-fall lesson (student)

Opens the “Gravity and free fall” lesson (ED-4), drops two spheres of different mass (FR-P2), sees both land together with velocity vectors shown (ED-3), reads $v = gt$ with live values (ED-7).

UC-3 — Projectile experiment (teacher in class)

Sets up a launcher scene, applies an impulse at 30° then 45° (FR-P5), compares overlaid trajectories (ED-6), saves the scene as an exercise for students (ED-8).

UC-4 — Sculpt and print (hobbyist)

Starts from a sphere (FR-G4), sculpts a character head with symmetry on (FR-S1, FR-S4), exports to STL for 3D printing (FR-G7).

UC-5 — Stability check (architecture student)

Stacks extruded blocks into a structure, runs the simulation (FR-P1), watches whether it stays up, adjusts and reruns.

8 Constraints and assumptions

Constraints

- Technology: Java with JavaFX for UI and 3D viewport, Maven build
- Timeline: one academic project cycle; core modules must be demo-ready before stretch work starts.
- Team: small student team; no dedicated designer or physics specialist.
- Hardware: must run on typical student laptops, not gaming machines.
- JavaFX 3D has no built-in physics engine, so physics is implemented in-house or via a lightweight Java library.

Assumptions

- Users have a mouse or trackpad; touch and pen input are not required.
- Simulation accuracy needs to be pedagogically correct (right trends, right formulas), not engineering-grade.
- Teachers will distribute exercises as files; no server or accounts are needed.
- The existing prototype (sculpt brush, undo/redo, object list) is the starting codebase.

9 Prioritization and release plan

Work ships in four increments; each one is usable on its own, and stretch modules start only after increment 3 passes its acceptance criteria.

1. **Modelling core** — all FR-G Must items, sculpting (FR-S1–S3), Draw and Extrude as separate tools (FR-D1, FR-D3, FR-D4), OBJ/STL export.
2. **Physics core** — FR-P1 to FR-P4, plus force and velocity overlays (ED-3).
3. **Education layer** — Explain toggle, geometry overlays, first 3 guided lessons (ED-1, ED-2, ED-4), remaining Should items as time allows.
4. **Stretch** — animation (FR-A1, FR-A2) first, fluid simulation (FR-F1) only if time remains.

MoSCoW: Must items form the minimum viable product; Should items are expected for the final evaluation; Could items are bonus scope that can be cut without failing the project.

References

- [1] G2, *Blender Reviews*. https://www.g2.com/products/blender/reviews_and_filters
- [2] School of Motion, *Is Blender Hard to Learn?* <https://www.schoolofmotion.com/blog/is-blender-hard>